
A Median Perspective on Unlabeled Data for Out-of-Distribution Detection

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Abstract

1 Out-of-distribution (OOD) detection is central to the robustness and reliability
2 of machine learning systems deployed in the real world, yet the most readily
3 available signal for it—unlabeled “in-the-wild” data—is hard to exploit because it
4 mixes in-distribution (InD) and OOD samples with no labels telling the two apart.
5 This missing separation is precisely what blocks the training of an accurate OOD
6 classifier. This paper makes three contributions that together turn this liability
7 into a usable signal. **First**, we introduce *Medix*, a framework whose core is a
8 median-centric filtering stage that extracts likely outliers from unlabeled data; we
9 build on the median because it estimates central tendency stably and is provably
10 robust to noise and contamination, and we then train an OOD classifier on the
11 extracted outliers together with labeled InD data. **Second**, we place this filter on
12 firm theoretical ground, deriving two-sided error bounds that certify a low misclas-
13 sification rate for both retained inliers and flagged outliers. **Third**, we validate
14 these guarantees empirically: across open-world benchmarks *Medix* outperforms
15 existing methods across the board, directly confirming the theory. In short, the
16 median perspective is what makes unlabeled wild data both practically effective
17 and theoretically controllable for OOD detection.

1 Introduction

19 Deploying machine learning models in real-world applications repeatedly exposes them to out-of-
20 distribution (OOD) inputs that threaten safety and reliability. Such inputs come from unknown
21 categories the model was never meant to predict, yet neural networks are inherently vulnerable and
22 typically lack any built-in mechanism to detect and appropriately handle them in practice (Nguyen
23 et al., 2015). This gap is the problem our paper sets out to close.

24 The difficulty is that OOD samples cannot be identified reliably at inference time: the model never
25 sees unknown distributions during training and so cannot cleanly separate OOD from in-distribution
26 (InD) data. A promising line of work (Katz-Samuels et al., 2022a; Du et al., 2024a) sidesteps this by
27 leveraging additional “in-the-wild” data. Katz-Samuels et al. (2022a), for instance, use unlabeled
28 wild data to regularize training while still classifying labeled InD data. The appeal of wild data is
29 its sheer availability—it is easily collected once a model is deployed—and using it lets the model
30 capture the true test-time OOD distribution, yielding more robust detection.

31 The catch, and the specific obstacle we target, is that wild data is a complex, *unlabeled* mixture of
32 InD and OOD samples: the detector sees only the mixture and never learns which individual samples
33 are OOD. This missing separation is what makes a clean, robust detector hard to train, and the
34 problem remains underexplored. Compounding this, almost no prior work offers a formal theoretical
35 foundation for the in-the-wild setting; to our knowledge, Du et al. (2024a) stands alone in providing
36 one. A method that both recovers the missing separation *and* comes with guarantees is therefore what
37 the field is missing.

Our starting point is a recent observation that median-based approaches are highly effective for data pruning (Acharya et al., 2024). Building on this, we ask the question that organizes the entire paper:

How can median-based methods leverage unlabeled wild data to facilitate OOD detection with theoretical guarantees?

We answer this question affirmatively by introducing a median-centric perspective on OOD detection, and the answer takes the form of three tightly linked contributions—a method, a theory that certifies it, and experiments that confirm the theory. On the method side, we propose a median-based optimization framework and an algorithm, Medix, that extracts OOD samples from wild data at a low error rate. On the theory side—where we are among the very few to offer any guarantee for the unlabeled in-the-wild setting—we prove that median-based filtering is robust by bounding the fraction of InD samples wrongly flagged as outliers. That fraction is controlled by two interpretable effects that recur throughout the paper: a *contamination effect*, which measures the pull of OOD points and stays manageable as long as the OOD proportion is below 50%; and a *concentration effect*, which uses the sub-Gaussian behavior of InD gradients to bound how far InD points stray from their mean gradient. Together these ensure that, with high probability, few InD points are misclassified even in the worst case.

On the empirical side, we benchmark Medix against two families of methods: (1) those trained on InD data alone, and (2) those trained on InD data plus an auxiliary unlabeled dataset. On CIFAR-100 (Krizhevsky et al., 2009), Medix improves markedly over the strong KNN+ baseline (Sun et al., 2022), cutting FPR95 by an average of 40.98%. Crucially, unlike Outlier Exposure (Hendrycks et al., 2019), which assumes a clean auxiliary dataset fully separable from the InD data, Medix needs no such assumption and still wins, offering greater flexibility. Against WOODS (Katz-Samuels et al., 2022a), Medix lowers the average FPR95 by 1.32% on CIFAR-100 and 2.60% on CIFAR-10.

We summarize our three contributions—method (C1), theory (C2), and evaluation (C3)—as follows, and we return to them explicitly in the method, theory, experiments, and conclusion sections so the reader can track each claim to its evidence:

- C1) Method.** We propose Medix, a median-centric greedy framework that first filters outliers from unlabeled wild data and then trains an OOD detector on the identified outliers together with the InD samples. The filtering stage is our central methodological contribution.
- C2) Theory.** We establish two-sided theoretical guarantees for median-based filtering in unlabeled mixtures. Concretely, we prove that both error types—InD samples flagged as outliers and OOD samples retained as inliers—are tightly bounded by the same two effects introduced above: a contamination effect that stays controlled as long as the OOD proportion is below 50%, and a concentration effect that reflects the stability of InD gradient behavior.
- C3) Evaluation.** We evaluate Medix extensively across eleven InD–OOD pairs against 20 competitive baselines, and it outperforms all of them, achieving superior performance across the board. We further show that Medix’s outlier extraction attains a low error rate (as low as 12.5%; see Figure 2), directly corroborating the theory of C2.

2 Preliminaries and Problem Setup

Before presenting our contributions, we fix the setting they operate in. This section overviews the OOD detection problem and then formally defines the data setup, model architecture, loss functions, and learning goal—the ingredients that make the missing InD/OOD separation (and thus the need for C1’s filtering stage) precise.

Labeled In-Distribution Data. Let $\mathcal{X}$ be the input space and $\mathcal{Y} = \{1, \dots, K\}$ the label space, which together define the InD structure. The labeled dataset $\mathcal{S}_{\text{in}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ is drawn i.i.d. from $\mathbb{P}_{XY}$, an unknown joint distribution over $\mathcal{X} \times \mathcal{Y}$, whose marginal on $\mathcal{X}$ —the InD input distribution—we write $\mathbb{P}_{\text{in}}$. We train an InD model on $\mathcal{S}_{\text{in}}$; its gradients later supply the reference signal that C1 filters against.

Out-of-distribution detection. We target a practical scenario: the model is trained on labeled InD data but later deployed where inputs may be OOD, drawn from classes absent at training time, i.e.,

some label $y \notin \mathcal{Y}$. On such inputs the model should abstain. The core inference-time objective is therefore to decide whether a given input belongs to the InD distribution or comes from an OOD source.

Unlabeled wild data. A primary obstacle in OOD detection is the scarcity of labeled OOD samples: the OOD sample space is vast, so collecting labeled examples is costly and often impractical. Following Katz-Samuels et al. (2022a), we sidestep this by bringing unlabeled wild data, $\mathcal{S}_{\text{wild}} = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_m\}$, into the learning framework to better mimic real deployment. Wild data blends InD and OOD samples and is readily collected once a model pre-trained on $\mathcal{S}_{\text{in}}$ is deployed—making it exactly the unlabeled mixture that C1 is designed to disentangle. Following Du et al. (2024a); Katz-Samuels et al. (2022a), we model the marginal distribution of the wild data with the Huber contamination model (Huber, 1964):

$$\mathbb{P}_{\text{wild}} := (1 - \pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}, \quad \pi \in (0, 1], \quad (1)$$

where π is the contamination proportion and $\mathbb{P}_{\text{out}}$ is the OOD distribution over $\mathcal{X}$; $\pi = 0$ would mean no OOD samples at all, making the problem trivial.

Models and Loss Functions. The InD classifier $f_\phi : \mathcal{X} \rightarrow \mathbb{R}^K$ (parameters $\phi \in \Phi$) outputs a soft distribution over the $K = |\mathcal{Y}|$ InD classes, trained with loss $\ell : \mathbb{R}^K \times \mathcal{Y} \rightarrow \mathbb{R}$. Separately, the OOD detector $g_\theta : \mathcal{X} \rightarrow \mathbb{R}$ (parameters $\theta \in \Theta$) uses the binary loss $\ell_b(g_\theta(\mathbf{x}), y_b)$ with $y_b \in \mathcal{Y}_b := \{y_+, y_-\}$, where $y_+ > 0$ marks the InD class and $y_- < 0$ the OOD class. Both models reappear as the two objects trained in C1’s second stage.

Learning objective. Our framework simultaneously trains the OOD detector g_θ and the multi-class classifier f_ϕ , leveraging both the InD data $\mathbb{P}_{\text{in}}$ and the wild data $\mathbb{P}_{\text{wild}}$ —the dual use of these two sources is what C1 operationalizes. At test time, we report the following three metrics:

$$\begin{aligned} \downarrow \text{FPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{out}}^{\text{test}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{TPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{in}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{Acc}(f_\phi) &= \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{\mathcal{X}\mathcal{Y}}} [\mathbb{I}\{f_\phi(\mathbf{x}) = y\}], \end{aligned}$$

where $\mathbb{I}\{\cdot\}$ denotes the indicator function, and $\mathbb{P}_{\text{out}}^{\text{test}}$ represents the OOD test data distribution.

3 Method: Median-Centric Framework for OOD Detection

This section develops contribution C1: **Medix**, a learning paradigm that harnesses unlabeled wild data for OOD detection. Medix overcomes the limitation of conventional approaches that rely on InD data alone, and is tailored to open-world settings where models routinely face previously unseen inputs. It comprises two stages: **1) Outlier Extraction**—a filtering process that isolates candidate OOD samples from the unlabeled wild data (Section 3.1), and **2) Detector Training**—training a binary OOD detector on both InD data and the outlier candidates from stage 1 (Section 3.2), following the protocol of Du et al. (2024a). We stress that the filtering stage is where the novelty of C1 lies; stage 2 is deliberately standard so that any gain can be attributed to the median-centric extraction. As the following subsections show, this two-step design both extracts OOD data effectively from the unlabeled mixture and lays a robust foundation for deployment in dynamic, open-world scenarios.

3.1 Extracting Candidate Outliers from the Wild Data

The filtering stage—the heart of C1—isolates potential outliers from the wild mixture $\mathcal{S}_{\text{wild}}$ through an optimization-based approach built on model-parameter gradients. These gradients come from a classifier f_ϕ trained solely on the InD dataset $\mathcal{S}_{\text{in}}$. We formalize the procedure below in three steps: estimating a reference gradient, motivating the median criterion, and formulating the filtering objective.

Reference gradient estimation from InD data. The first step in our proposed framework is to estimate a reference gradient using the InD dataset $\mathcal{S}_{\text{in}}$. This is achieved by training a classifier f_ϕ on $\mathcal{S}_{\text{in}}$ through empirical risk minimization (ERM) as follows

$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi \in \Phi} \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi), \quad \text{where } \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi) = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_\phi(\mathbf{x}_i), y_i), \quad (2)$$

where $\phi_{S_{\text{in}}}$ denotes the learned parameters. Once the classifier has been trained, we compute the mean gradient $\bar{\nabla}_{\text{in}}$ as the average of the gradients of the loss function with respect to the model parameters over the InD data:

$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in S_{\text{in}}} \nabla \ell(f_{\phi_{S_{\text{in}}}}(\mathbf{x}_i), y_i). \quad (3)$$

This $\bar{\nabla}_{\text{in}}$ is our reference gradient: it lets us quantify how far any other data point deviates from typical InD behavior, and every subsequent filtering decision is measured against it.

Motivation. The design of C1 rests on a simple, testable hypothesis: adding more OOD samples to the wild dataset S_{wild} should push its gradient statistics farther from the average InD gradient $\bar{\nabla}_{\text{in}}$. We test this directly, using CIFAR-10 (Krizhevsky et al., 2009) as InD and SVHN (Netzer et al., 2011) as OOD. Here S_{wild} holds 10,000 CIFAR-10 samples disjoint from the set used to train $\phi_{S_{\text{in}}}$ (from which we compute $\bar{\nabla}_{\text{in}}$). We then incrementally inject SVHN OOD samples and track the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median (EWM) of the wild gradients:

$$\left\| \bar{\nabla}_{\text{in}} - \text{EWM} \left(\left\{ \nabla \ell \left(f_{\phi_{S_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in S_{\text{wild}}} \right) \right\|.$$

Figure 1 confirms the hypothesis: the L_2 -norm deviation rises clearly and monotonically with the OOD count. This single observation does double duty for C1—it motivates the median-based filter and it directly supplies its stopping criterion, since we halt once the L_2 -norm deviation between consecutive iterations falls below a threshold ϵ . We detail this next.

Filtering potential outliers from unlabeled wild data. Motivated by Figure 1, we cast C1’s filtering step as the following optimization problem, whose solution identifies the outlier subset S_{out}^* in S_{wild} :

$$S_{\text{in}}^* = \arg \min_{S \subseteq S_{\text{wild}}} \left\| \bar{\nabla}_{\text{in}} - \text{EWM}(G_S) \right\|, \text{ where } G_S = \left\{ \nabla \ell \left(f_{\phi_{S_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in S}. \quad (4)$$

Here $\text{EWM}(\cdot)$ denotes the element-wise median function, and $\hat{y}_{\tilde{\mathbf{x}}_i}$ represents the predicted label for a wild sample $\tilde{\mathbf{x}}_i$. For notational simplicity, we define $G_S = \{\nabla \ell(\tilde{\mathbf{x}}_i)\}_{i \in S}$. Intuitively, this objective seeks the subset S of S_{wild} whose gradient EWM is closest to the average InD gradient $\bar{\nabla}_{\text{in}}$. By the trend in Figure 1, the minimizer S_{in}^* should recover the InD portion of S_{wild} , so its complement $S_{\text{out}}^* = S_{\text{wild}} \setminus S_{\text{in}}^*$ captures the OOD portion—precisely the separation the field was missing.

Solving (4) exactly is computationally prohibitive as S_{wild} grows. C1 therefore contributes a tractable greedy approximation based on a leave-one-out rule, given in Algorithm 1. The algorithm iteratively detects outliers by tracking gradient deviations relative to the InD reference.

The algorithm begins by computing the EWM of the wild data gradients at each iteration, which serves as a reference for comparing against the average gradient of the InD data $\bar{\nabla}_{\text{in}}$. We denote by d_t , the L_2 distance between the average InD gradient $\bar{\nabla}_{\text{in}}$ and the EWM gradients of the data left in the wild set, represented by S , i.e., $d_t = \|\text{EWM}(G_S) - \bar{\nabla}_{\text{in}}\|$. The algorithm then iteratively identifies samples in S that incur the most significant drop in the L_2 distance with d_t when removed from S as OOD. Specifically, having data samples S left, we remove each sample $i \in S$ and compute the EWM as $\text{EWM}(G_{S \setminus \{i\}})$ and the distance to $\bar{\nabla}_{\text{in}}$ as $\|\text{EWM}(G_{S \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$. We then find the drop in distance $\delta_i = d_t - \|\text{EWM}(G_{S \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$ and find the k samples with the largest drop and identify them as OOD. The algorithm repeats until there is no significant drop in δ_i or a maximum number of iterations is reached. Convergence is decided by the change in L_2 distance across iterations: once it drops below the ϵ threshold, we stop. This rule follows directly from the monotone trend of Figure 1—as OOD samples are removed, the L_2 distance shrinks to a small value, marking the moment to halt before InD samples are mistakenly flagged, which would otherwise degrade performance.

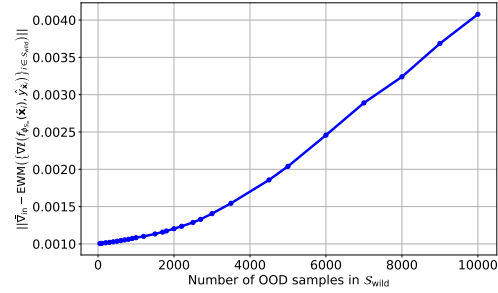


Figure 1: Motivation for C1: the L_2 distance deviation grows monotonically as more OOD samples enter S_{wild} , justifying both the median criterion and the stopping rule.

Algorithm 1 Iterative Outlier Detection via **Medix**

Require: $\bar{\nabla}_{\text{in}}, G_i = \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i})}$, maximum iterations T , hyperparameters ϵ, k

- 1: Initialize $\mathcal{S} \leftarrow \mathcal{S}_{\text{wild}}$ (wild set), $\mathcal{O} \leftarrow \emptyset$ (outliers), $t \leftarrow 0$ (iteration), $d_t \leftarrow 0$, $\delta_{\text{max}} \leftarrow \infty$ (deviations), $\mathcal{I}_k \leftarrow \emptyset$
- 2: **while** $t \leq T$ or $|\delta_{\text{max}}| > \epsilon$ **do**
- 3: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\tilde{\mathbf{x}}_i : i \in \mathcal{I}_k\}$ ▷ Add outliers to set $\mathcal{O}$
- 4: $d_t \leftarrow \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$ ▷ Compute L2 deviation
- 5: **for each** $i \in \mathcal{S}$ **do**
- 6: $\delta_i \leftarrow d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$
- 7: **end for**
- 8: $\mathcal{I}_k \leftarrow \text{indices}(\text{top-}k(\{\delta_i\}_{i \in \mathcal{S}}))$ ▷ Select top- k indices
- 9: $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}_k$ ▷ Remove outliers from $\mathcal{S}$
- 10: $\delta_{\text{max}} \leftarrow \max_{i \in \mathcal{S}} \{\delta_i\}$, $t \leftarrow t + 1$
- 11: **end while**
- 12: **return** $\mathcal{S}_{\text{out}} = \mathcal{O}$ ▷ Return detected outliers

3.2 Training the OOD Detector with Candidate Outliers

With the candidate outlier set $\hat{\mathcal{S}}_{\text{out}}$ recovered by C1’s filter, stage 2 trains the OOD detector g_{θ} to separate InD from candidate-outlier data, following the protocol of Du et al. (2024a). We keep this stage intentionally standard so that the paper’s gains trace back to the filtering contribution rather than to detector-side novelty. The objective enforces separability at the decision boundary (thresholded at 0), yielding positive outputs for labeled InD samples $\mathbf{x} \in \mathcal{S}_{\text{in}}$ and negative outputs for candidate outliers $\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}$:

$$\begin{aligned} \mathcal{L}_{\mathcal{S}_{\text{in}}, \hat{\mathcal{S}}_{\text{out}}}(g_{\theta}) &= \mathcal{L}_{\mathcal{S}_{\text{in}}}^{+}(g_{\theta}) + \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^{-}(g_{\theta}), \\ \mathcal{L}_{\mathcal{S}_{\text{in}}}^{+}(g_{\theta}) &= \mathbb{E}_{\mathbf{x} \in \mathcal{S}_{\text{in}}} \mathbb{I}\{g_{\theta}(\mathbf{x}) \leq 0\}, \\ \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^{-}(g_{\theta}) &= \mathbb{E}_{\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}} \mathbb{I}\{g_{\theta}(\tilde{\mathbf{x}}) > 0\}. \end{aligned} \tag{5}$$

Because the 0/1 loss is non-differentiable, we optimize a smooth sigmoid-based surrogate that keeps the problem tractable while staying aligned with the original objective. The framework also folds in the InD risk of Equation 2, safeguarding InD predictive accuracy. This unified objective sharpens the generalization of g_{θ} and enables robust detection of OOD samples from $\mathbb{P}_{\text{out}}$ —a benefit our experiments (C3) confirm directly.

4 Theoretical Analysis

This section delivers contribution C2: theoretical guarantees for Medix’s filtering stage. The two theorems below provide provable upper bounds on the misclassification rates for InD and OOD points respectively, and jointly establish the two-sided robustness of EWM filtering that we claimed in the introduction. This is what sets Medix apart in a setting where formal guarantees are almost entirely absent. Detailed proofs appear in Appendix C.

Theorem 4.1 (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the EWM filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

Theorem 4.1 is the first half of C2: it shows the inlier misclassification rate is governed by exactly the two effects previewed in the introduction—a *concentration term* that vanishes as the sample size grows and captures statistical fluctuation in the InD gradients, and a *contamination term* that measures how OOD points pull on the EWM. Critically, the bound stays controlled whenever the contamination

ratio $\pi < 0.5$, formalizing the robustness of median-based filtering to moderate OOD presence. This settles one direction; the complementary question—*how many true outliers slip through and are mistakenly retained?*—is answered by the next theorem, completing the two-sided guarantee.

Theorem 4.2 (Outlier Misclassification Bound). *Assume that the gradients of OOD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ_{out}^2 . Suppose the mean OOD gradient μ_{out} satisfies a separation condition:*

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d},$$

for some $\Delta > 0$. Then, for any tolerance $\epsilon \in (0, \Delta)$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the outlier misclassification rate satisfies:

$$\text{ERR}_{\text{out}} \leq \underbrace{2d \exp\left(-\frac{(\Delta - \epsilon)^2}{2\sigma_{\text{out}}^2}\right)}_{\text{Separation term}} + \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}}_{\text{Concentration term}} + \underbrace{\frac{1 - \pi}{2\pi}}_{\text{Contamination term}}.$$

Theorem 4.2 supplies the second half of C2, bounding the fraction of OOD points mistakenly kept as InD. Taken together, Theorems 4.1 and 4.2 deliver the full two-sided robustness guarantee promised in C2. Both misclassification rates are controlled by a balance among the same three interpretable effects, which we unpack in turn:

Contamination Effect. In Theorem 4.1, the median criterion stays stable even under a substantial OOD fraction π , provided $\pi < 0.5$. This is exactly the $\pi/[2(1 - \pi)]$ term, which grows with π and encodes the risk of OOD gradients skewing the EWM. Theorem 4.2 carries the symmetric penalty $\frac{1 - \pi}{2\pi}$, quantifying how hard OOD points are to isolate when they are underrepresented. This is the same contamination effect flagged in C2.

Concentration Effect. Both bounds exploit the sub-Gaussian nature of the gradient coordinates. For InD data, this keeps most gradients tightly around $\bar{\nabla}_{\text{in}}$, stabilizing the median even in finite samples; for OOD data, concentration around μ_{out} lets us bound the chance that a misaligned OOD sample slips past. Both terms decay as $1/\sqrt{m}$ in the per-class sample size m , so more data tightens C2’s guarantee.

Separation Effect. Unique to Theorem 4.2, this effect states how far the OOD mean gradient must sit from the InD mean gradient for reliable rejection. The exponential term $\exp(-(\Delta - \epsilon)^2/2\sigma_{\text{out}}^2)$ captures the trade-off: the more separated the distributions, the less likely OOD gradients fool the filter.

Takeaway (C2). Theorems 4.1 and 4.2 jointly certify that, with high probability, median filtering robustly separates InD and OOD samples in unlabeled mixtures. The InD-as-outlier error is bounded by contamination and concentration effects; the OOD-as-inlier error is bounded by an exponential separation term, a concentration bound, and a reverse contamination effect. Under mild assumptions, Medix provably keeps both error types small—the rigorous foundation this paper set out to establish.

Remark 4.3. Sub-Gaussian Assumption for Gradient Coordinates. The one assumption C2 leans on—that each coordinate of the InD gradients is sub-Gaussian—is what makes the concentration bounds go through, and we verify it empirically rather than take it on faith. Figure 4a shows the InD gradient histogram is bell-shaped and tightly concentrated around the mean, indicating the light, exponentially decaying tails that define sub-Gaussian variables (extreme gradients are rare). Figure 4b reinforces this via a Q-Q plot (Wilk and Gnanadesikan, 1968): the empirical gradient quantiles hug the 45-degree Gaussian reference line, and since Gaussians are sub-Gaussian, the assumption holds. *To show the guarantee is not fragile to this choice, we also prove a looser version that removes the sub-Gaussian assumption entirely, needing only bounded second moments (Theorem C.3, Appendix C.3). The rates degrade, but Medix’s core robustness guarantee survives—broadening the reach of C2.*

5 Experiments

This section delivers contribution C3: an extensive empirical validation of Medix across InD–OOD dataset pairs against 20 widely used baselines. Our goal is twofold—to show that Medix wins across the board, and to confirm the low-error-rate prediction of the theory (C2) in practice. All experiments run on NVIDIA A100-SXM4-80GB GPUs, and code to reproduce our results will be released upon acceptance.

5.1 Models, Datasets, and Baselines

Datasets. For a fair comparison with prior work, we adopt the exact protocol of Katz-Samuels et al. (2022a), which introduced the problem of learning OOD detectors from wild data. We use CIFAR-10 and CIFAR-100 as InD datasets ($\mathbb{P}_{\text{in}}$), and a suite of natural-image OOD datasets ($\mathbb{P}_{\text{out}}$): PLACES365 (Zhou et al., 2017), SVHN, TEXTURES (Cimpoi et al., 2014), and LSUN-RESIZE & LSUN-C (Yu et al., 2015). To simulate wild data ($\mathbb{P}_{\text{wild}}$), we mix a subset of InD data ($\mathbb{P}_{\text{in}}$) with OOD data ($\mathbb{P}_{\text{out}}$) at a default contamination $\pi = 0.5$. For PLACES365, say, the wild mixture combines CIFAR with PLACES365, and we test on PLACES365 as the OOD set; we repeat this for every OOD dataset and baseline. The InD CIFAR data is split in half—the first 25,000 images train $\phi_{\mathcal{S}_{\text{in}}}$, and the rest form the wild mixture $\mathcal{S}_{\text{wild}}$. Gradients are computed from the penultimate-layer weights, which are known to be especially informative for OOD detection (Huang et al., 2021a).

Evaluation metrics. We report three standard metrics: (1) the False Positive Rate of OOD samples at 95% InD True Positive Rate (FPR95 $\downarrow$), (2) the Area Under the ROC Curve (AUROC $\uparrow$), and (3) InD Classification Accuracy (InD Acc $\uparrow$).

Baselines. To make the C3 comparison as demanding as possible, we span a diverse set of competitive baselines, grouped by whether they use InD data only or both InD and wild data. Among methods trained on InD data alone ($\mathbb{P}_{\text{in}}$), we compare against established detectors—Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2016), ODIN (Liang et al., 2018), Mahalanobis Distance (Lee et al., 2018a), Energy Score (Liu et al., 2020b), ReAct (Sun et al., 2021), DICE (Sun and Li, 2022), KNN Distance (Sun et al., 2022), and ASH (Djurisic et al., 2022), all built on a softmax cross-entropy model—and, for breadth, against contrastive-loss methods CSI (Tack et al., 2020a) and KNN+ (Sun et al., 2022). Among methods that use both InD and wild data, we compare against Outlier Exposure (OE) (Hendrycks et al., 2019) and energy-regularization learning (Liu et al., 2020b), which regularize training toward lower confidence or higher energy on outliers; WOODS (Katz-Samuels et al., 2022a), which introduced wild unlabeled data and exploits it via constrained optimization; and the recent CONJ (Peng et al., 2024) and DRL (Zhang et al., 2024), added for a still more thorough evaluation.

5.2 Experimental Setup

Following WOODS (Katz-Samuels et al., 2022a), the InD classifier $\phi_{\mathcal{S}_{\text{in}}}$ is a 40-layer Wide ResNet (Zagoruyko, 2016) with width factor 2, trained by SGD (momentum 0.9, weight decay 0.0005, initial learning rate 0.1) for 100 epochs with cosine decay, batch size 128, and dropout 0.3. The two Medix hyperparameters ϵ and k are chosen from $\{5\text{e-}5, 5\text{e-}4, 5\text{e-}3, 5\text{e-}2\}$ and $\{4\text{k}, 7\text{k}, 10\text{k}, 20\text{k}\}$ respectively, based on dataset size and OOD performance—a search we later show (C3, Appendix A.2) Medix is largely insensitive to. The OOD classifier g_{θ} is initialized from the pre-trained $\phi_{\mathcal{S}_{\text{in}}}$ with an added linear layer doing binary classification on penultimate-layer features; it is fine-tuned for 100 epochs at learning rate 0.001 per Equation (5). We combine the binary loss with the InD classification loss, weighting the binary term by 10, and keep all other settings identical to those used for $\phi_{\mathcal{S}_{\text{in}}}$.

5.3 Results

Our main results, in Table 2 on CIFAR-100, deliver the central claim of C3: Medix outperforms every OOD detection baseline. Two observations stand out. (1) *Wild data helps, and Medix uses it best.* Methods trained on both InD and wild data clearly beat InD-only methods, and Medix cuts FPR95 by 52.31% on PLACES365 and 38.24% on TEXTURES relative to KNN+, confirming the value of in-the-wild regularization. (2) *Medix beats other wild-data methods too.* It reaches an average FPR95 of 5.42% on CIFAR-100—a 1.32% improvement over WOODS—while keeping a competitive InD

Table 1: Evidence for C3 on CIFAR-10 (InD): OOD detection performance of Medix versus all baselines. Medix (shaded) attains the best FPR95 and AUROC on every OOD set. Results are averaged over five runs; best values are in **bold**.

Methods	OOD Datasets												InD Acc $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
Using $\mathbb{P}_m$ only													
MSP	48.49	91.89	59.48	88.20	30.80	95.65	52.15	91.37	59.28	88.50	50.04	91.12	94.84
ODIN	33.35	91.96	57.40	84.49	15.52	97.04	26.62	94.57	49.12	84.97	36.40	90.61	94.84
Mahalanobis	12.89	97.62	68.57	84.61	39.22	94.15	42.62	93.23	15.00	97.33	35.66	93.34	94.84
Energy	35.59	90.96	40.14	89.89	8.26	98.35	27.58	94.24	52.79	85.22	32.87	91.73	94.84
KNN	24.53	95.96	25.29	95.69	25.55	95.26	27.57	94.71	50.90	89.14	30.77	94.15	94.84
ReAct	89.57	81.44	94.44	90.44	14.38	97.21	33.63	93.58	53.63	86.59	36.77	91.48	94.84
DICE	35.44	89.65	46.83	86.69	6.32	98.68	28.93	93.56	53.62	82.20	34.23	90.16	94.84
ASH	6.51	98.65	48.45	88.34	0.90	99.73	4.96	98.92	24.34	95.09	17.03	96.15	94.84
CSI	17.30	97.40	34.95	93.64	1.95	99.55	12.15	98.01	20.45	95.93	17.36	96.91	94.17
KNN+	2.99	99.41	24.69	94.84	2.95	99.39	11.22	97.98	9.65	98.37	10.30	97.99	93.19
Using $\mathbb{P}_m$ and $\mathbb{P}_{wild}$													
OE	1.13	99.53	19.48	94.88	1.91	98.16	0.54	98.84	7.75	98.56	6.16	97.99	94.12
Energy (w/ OE)	5.24	98.72	14.66	96.18	2.35	99.30	4.85	98.62	10.51	97.10	7.52	97.98	94.24
WOODS	0.17	99.91	10.19	98.05	0.31	99.14	0.11	99.38	6.21	98.13	3.40	98.92	94.74
Medix	0.06	99.98	2.98	99.10	0.01	99.98	0.01	99.98	0.96	99.66	0.80	99.74	93.58
(Ours)	± 0.01	± 0.01	± 0.29	± 0.09	± 0.01	± 0.00	± 0.01	± 0.01	± 0.13	± 0.06	± 0.09	± 0.03	± 0.64

Table 2: Evidence for C3 on CIFAR-100 (InD): OOD detection performance of Medix versus all baselines. Medix (shaded) again leads on FPR95 and AUROC across the board. Results are averaged over five runs; best values are in **bold**.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
Using $\mathbb{P}_m$ only													
MSP	84.59	71.44	82.84	73.78	66.54	83.79	82.42	75.38	83.29	73.34	79.94	75.55	75.96
ODIN	84.66	67.26	87.88	71.63	55.55	87.73	71.96	81.82	79.27	73.45	75.86	76.38	75.96
Mahalanobis	57.52	86.01	88.83	67.87	91.18	69.69	21.23	96.00	39.39	90.57	59.63	82.03	75.96
Energy	85.82	73.99	80.56	75.44	35.32	93.53	79.47	79.23	79.41	76.28	72.12	79.69	75.96
KNN	66.38	83.76	79.17	71.91	70.96	83.71	77.83	78.85	88.00	67.19	76.47	77.08	75.96
ReAct	88.04	81.33	74.32	39.30	91.19	79.86	73.69	67.38	82.80	68.44	82.01	75.96	
DICE	88.35	72.58	81.61	75.07	26.77	94.74	80.21	78.50	76.29	76.07	70.65	79.39	75.96
ASH	21.36	94.28	68.37	71.22	15.27	95.65	68.18	85.42	40.87	92.29	42.81	87.77	75.96
CSI	64.70	84.97	82.25	73.63	38.10	92.52	91.55	63.42	74.70	92.66	70.26	81.44	69.90
KNN+	32.21	93.74	68.30	75.31	40.37	86.13	44.86	88.88	46.26	87.40	46.40	86.29	73.78
Using $\mathbb{P}_m$ and $\mathbb{P}_{wild}$													
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17	73.38
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51	72.76
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73	73.91
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96	73.33
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10	± 0.83

accuracy of 73.33%. The small accuracy gap simply reflects that Medix trains on 25,000 labeled InD samples, whereas InD-only baselines use the full 50,000-image CIFAR-100 training set. The CIFAR-10 results in Table 1 tell the same story: Medix surpasses all baselines, improving FPR95 over WOODS by 7.21% on PLACES365 and 5.25% on TEXTURES. Together, both tables substantiate C3’s “across-the-board” claim on the standard benchmarks.

A representative visual example of Medix. To make C3’s connection to the theory tangible, we visualize the outlier-extraction step (Algorithm 1) on a deliberately simple 2-dimensional synthetic setup. InD data are drawn from three multivariate Gaussians (three classes) with means $[-2, 0]$, $[2, 0]$, and $[0, 2\sqrt{3}]$ and shared covariance $0.25 \cdot I$; OOD data come from $\mathcal{N}([20, 2\sqrt{3}], 0.25 \cdot I)$. As Figure 2 shows, Medix flags outliers that overwhelmingly match the ground truth—recovering 87.5% of true OOD samples—so the empirically observed error rate lands exactly where C2’s bounds say it should.

Additional studies that reinforce C3. Space limits force us to defer further experiments to Appendix A, but each one strengthens a facet of C3: (1) hyperparameter ablation and sensitivity analysis (Appendix A.2), showing Medix is strongly robust to its choice of ϵ and k ; (2) an EWM-vs-geometric-median comparison (Appendix A.1), showing EWM is more sensitive to distributional shift and thus the more reliable filter; (3) a comprehensive comparison against methods with competitive contrastive objectives (Appendix A.3), where Medix still wins by a wide margin; (4) large-scale evaluation under the harder unseen-OOD setting $\mathbb{P}_{out}^{test} \neq \mathbb{P}_{out}$ (Appendix A.4), where Medix again leads substantially; (5) a computation- and memory-efficiency study (Appendix A.6); (6) an analysis of pseudo-label quality (Appendix A.5), showing resilience to noisy or low-confidence labels; and (7) a comparison with semi-supervised open-set recognition methods (Appendix A.7).

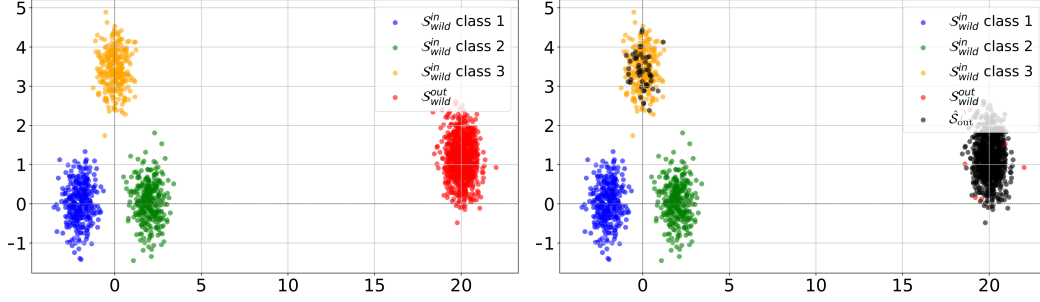


Figure 2: Visual confirmation that C1’s filter attains the low error rate predicted by C2. (a) The wild setup: InD data $S_{\text{wild}}^{\text{in}}$ (inliers from three multivariate Gaussians) and OOD data $S_{\text{wild}}^{\text{out}}$. (b) Outliers $\hat{S}_{\text{out}}$ flagged by Medix (black); only 12.5% of $\hat{S}_{\text{out}}$ are actually InD, i.e., an inlier-misclassification rate of just 12.5%.

6 Related Work

OOD detection has attracted intense recent interest (Fort et al., 2021; Yang et al., 2024; Fang et al., 2022; Zhu et al., 2022; Yang et al., 2022; Wang et al., 2022c; Galil et al., 2023; Djuriscic et al., 2023; Tao et al., 2023; Zheng et al., 2023; Wang et al., 2022b, 2023b; Uppaal et al., 2023; Zhu et al., 2023; Bai et al., 2023; Ming and Li, 2024; Zhang et al., 2023; Ghosal et al., 2024; Abbas et al., 2025). One family scores how distributional a sample is, via distance-based methods (Lee et al., 2018b; Tack et al., 2020b; Ren et al., 2021; Sehwag et al., 2021; Sun et al., 2022; Du et al., 2022a; Ming et al., 2023; Ren et al., 2022), gradient-based scores (Huang et al., 2021b), energy-based scores (Liu et al., 2020c; Wang et al., 2021; Wu et al., 2023), confidence-based approaches (Bendale and Boult, 2016; Hendrycks and Gimpel, 2017; Liang et al., 2018), and Bayesian methods (Gal and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox et al., 2019; Malinin and Gales, 2019; Wen et al., 2020; Kristiadi et al., 2020). Unlike these, Medix does not merely score inputs—it recovers an explicit outlier set from unlabeled data (C1).

A second family regularizes training (Malinin and Gales, 2018; Geifman and El-Yaniv, 2019; Hein et al., 2019; Meinke and Hein, 2020; Jeong and Kim, 2020; Liu et al., 2020a; Van Amersfoort et al., 2020; Yang et al., 2021; Wei et al., 2022; Du et al., 2022b, 2023; Wang et al., 2023a), for example by lowering confidence (Lee et al., 2017; Hendrycks et al., 2019) or raising energy (Liu et al., 2020c; Du et al., 2022c; Ming et al., 2022) on OOD data. Most such methods presuppose a clean auxiliary OOD dataset—an assumption C1 explicitly avoids by working directly on the unlabeled wild mixture.

Several studies (Zhou et al., 2021; Katz-Samuels et al., 2022b; He et al., 2023) relax the labeled-only assumption by incorporating unlabeled wild data (Katz-Samuels et al., 2022a; Geng et al., 2025), but propose no clear outlier-detection mechanism—precisely the gap C1 fills. Closer to us, Du et al. (2024a,b) introduce explicit outlier filtering, yet their thresholding differs fundamentally from our median-centric criterion. A further distinction is that Katz-Samuels et al. (2022a); Du et al. (2024a) assume batch-level mixing with a fixed InD/OOD ratio per batch; with large outsourced datasets no such structure exists—data is mixed randomly—and Medix handles this by operating at the dataset level rather than the batch level. Finally, positive-unlabeled learning trains classifiers from positive and/or unlabeled data (Letouzey et al., 2000; Hsieh et al., 2015; Plessis et al., 2015; Niu et al., 2016; Gong et al., 2018; Chapel et al., 2020; Garg et al., 2021; Xu and Denil, 2021; Garg et al., 2022; Zhao et al., 2022; Acharya et al., 2022), but only separates $\mathbb{P}_{\text{out}}$ from $\mathbb{P}_{\text{in}}$ without also training an OOD classifier. Against all of this, our combined offering is distinctive: a median-centric outlier identifier for unlabeled data (C1) that comes with provably low error rates (C2).

7 Conclusions

This paper introduced Medix, a median-centric framework that turns unlabeled in-the-wild data into a usable signal for OOD detection, and its value rests on the three contributions we set out to deliver. For C1 (method), Medix uses the inherent robustness of the median to filter outliers from a mixed unlabeled set and then trains a reliable OOD detector on them. For C2 (theory), we proved two-sided

338 bounds—on the inlier misclassification rate, robust even under up to 50% OOD contamination via
 339 sub-Gaussian concentration and contamination effects, and a complementary bound on the OOD
 340 misclassification rate under clear separation. For C3 (evaluation), Medix outperformed 20 baselines
 341 across diverse benchmarks, cutting average FPR95 by 40.98% relative to strong baselines such
 342 as KNN+, beating state-of-the-art methods including WOODS and DRL, and attaining an outlier-
 343 extraction error rate as low as 12.5%—exactly the low-error regime the theory predicts. Together,
 344 C1–C3 show that a median perspective makes unlabeled wild data both effective and theoretically
 345 controllable for OOD detection.

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606 A Additional Studies

607 A.1 Element-wise Median vs Geometric Median

608 A natural question about C1 is why we chose the element-wise median over the geometric
 609 median (Acharya et al., 2024) as Medix’s filtering operator; this study answers it directly.
 610 We ran a controlled experiment with CIFAR-10 as InD
 611 and SVHN as OOD. Here $\mathcal{S}_{\text{wild}}$ holds 5k CIFAR-10 sam-
 612 ples disjoint from the set used to train $\phi_{\mathcal{S}_{\text{in}}}$ (from which
 613 we compute $\bar{\nabla}_{\text{in}}$). We incrementally inject SVHN OOD
 614 samples and track both the L_2 -norm deviation between
 615 $\bar{\nabla}_{\text{in}}$ and the element-wise median of the wild gradients
 616 and the proportion of OOD samples removed. As Figure 3
 617 shows, at a matched OOD count the element-wise median
 618 flags a substantially larger fraction of true OOD samples
 619 than the geometric median. It is therefore more sensitive
 620 to distributional shift and the more effective, reliable filter
 621 for C1—which is exactly why Medix adopts it.

622 A.2 Hyperparameter 623 Selection and Sensitivity Analysis

624 To probe how much C1 depends on tuning, we ablate the
 625 two hyperparameters ϵ and k using CIFAR-100 as InD and
 626 SVHN as OOD. Table 3 shows Medix is strongly robust to both. At $\epsilon = 0.005$, $k = 20000$ it reaches
 627 a remarkably low FPR95 of 0.16, and perturbing the settings—dropping ϵ to 0.0005 or halving k to
 628 10000—still leaves it competitive (FPR95 of 0.20 and 0.68), a graceful degradation rather than a cliff.
 629 Across all these choices Medix continues to surpass the baselines. This insensitivity to exact tuning
 630 is a practical strength of C1: Medix stays reliable in real deployments where exhaustive tuning is
 631 infeasible.

Table 3: Robustness of C1 to its hyper-
parameters: FPR95 stays low as ϵ and
 k vary, and Medix beats the baselines
throughout.

Method	FPR95↓	ϵ	k
DICE	88.35	—	—
ASH	21.36	—	—
CSI	64.70	—	—
KNN+	32.21	—	—
OE	2.86	—	—
Energy	2.71	—	—
Medix	0.16	0.005	20000
Medix	0.20	0.0005	20000
Medix	0.68	0.005	10000

632 A.3 Comprehensive comparison with recent competitive methods

633 To stress-test C3 against the strongest available competition, we extend the comparison to baselines
 634 with diverse contrastive objectives—SSD+ (Sehwag et al., 2021), ProxyAnchor (Kim et al., 2020),
 635 and CIDER Ming et al. (2023)—which we include despite not using contrastive learning ourselves.
 636 We further add the recent CONJ (Peng et al., 2024), DRL (Zhang et al., 2024), Vim (Wang et al.,
 637 2022a), and VOS (Du et al., 2022c) for a still more thorough evaluation. For a fair comparison, we
 638 use CIFAR-100 as InD and train the ResNet-34 model¹ following Ming et al. (2023); Zhang et al.
 639 (2024): SGD with momentum 0.9 and weight decay 10^{-4} for 500 epochs, initial learning rate 0.5
 640 with cosine scheduling, and batch size 512. We reuse the checkpoints provided by Ming et al. (2023)².

¹Since ResNet-34 is the standard model across these works, we adopt the same model to ensure consistency
and facilitate a fair comparison.

²<https://github.com>

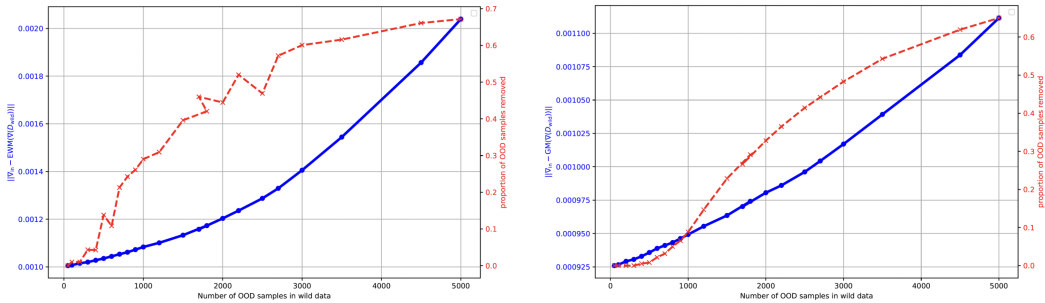


Figure 3: Supporting C3: element-wise median (EWM) versus geometric median (GM). For a
matched OOD count, EWM flags a larger fraction of true outliers, motivating our EWM choice in C1.

Table 4: Extending C3 to recent competitive baselines on CIFAR-100 (InD), including contrastive-objective methods. Medix (shaded) still leads on FPR95 and AUROC. Results are averaged over five runs; best values in **bold**.

Methods	OOD Datasets									
	SVHN		PLACES365		LSUN-C		TEXTURES		AVERAGE	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MSP	78.89	79.80	84.38	74.21	83.47	75.28	86.51	72.53	83.31	75.46
Mahalanobis	87.09	80.62	84.63	73.89	84.15	79.43	61.72	84.87	79.40	79.70
ODIN	70.16	84.88	82.16	75.19	76.36	80.10	85.28	75.23	78.49	78.85
Energy	66.91	85.25	81.41	76.37	59.77	86.69	79.01	79.96	71.78	82.07
ReAct	50.93	88.75	83.55	73.10	64.02	80.31	64.40	81.95	65.72	81.03
KNN	46.25	90.39	82.08	75.44	60.85	85.61	62.39	83.95	62.89	83.85
Vim	73.42	84.62	85.34	69.34	86.96	69.74	74.56	76.23	80.07	74.98
VOS	43.24	82.80	76.85	78.63	73.61	84.69	57.57	87.31	62.82	83.36
CSI	44.53	92.65	79.08	76.27	75.58	83.78	61.61	86.47	65.20	84.79
ProxyAnchor	87.21	82.43	70.10	79.84	37.19	91.68	65.64	84.99	65.04	84.74
SSD+	31.19	94.19	77.74	79.90	79.39	85.18	66.63	86.18	63.74	86.36
KNN+	39.23	92.78	80.74	77.58	48.99	89.30	57.15	88.35	56.53	87.00
ASH	52.96	90.19	72.62	76.38	75.18	76.52	56.17	86.75	64.23	82.46
CIDER	23.09	95.16	79.63	73.43	16.16	96.33	43.87	90.42	40.69	88.84
CONJ	46.19	90.44	80.81	75.83	60.45	85.90	62.13	83.77	62.40	83.99
DRL	20.15	94.07	76.64	77.55	16.97	94.63	31.97	92.09	36.43	89.59
Medix	0.48	99.87	24.52	93.42	0.73	99.84	8.99	97.92	8.68	97.76
(Ours)	± 0.05	± 0.02	± 1.76	± 0.36	± 0.08	± 0.03	± 0.72	± 0.24	± 0.65	± 0.16

Table 4 confirms the result: Medix outperforms even these strong baselines—contrastive methods included—by a wide margin, reaching an average FPR95 of 8.68% and AUROC of 97.76%. Beyond raw performance, Medix carries the provably low error rate of C2, a theoretical guarantee most of these alternatives lack—reinforcing that C1–C3 are complementary rather than redundant.

A.4 Evaluation on Large-Scale, Complex Unseen OOD Settings

We now push C3 into a harder regime: large-scale wild OOD data under the unseen-OOD setting. Two factors raise the difficulty: (1) the wild OOD dataset ($\mathbb{P}_{\text{out}}$) is large, increasing both data complexity and compute, and (2) the test-time OOD distribution ($\mathbb{P}_{\text{out}}^{\text{test}}$) is deliberately different from the training-time wild OOD data, i.e., $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$ —a more realistic and demanding deployment scenario. We use CIFAR-100 as labeled InD data, the 300K Random Images dataset Hendrycks et al. (2019) as unlabeled wild OOD data, and SVHN as the test OOD data—so the large-scale wild OOD (300K Random Images) differs sharply from the OOD test set (SVHN). Comparing against baselines that also use wild data in training, Table 5 shows Medix clearly on top, with a much lower FPR95 (41.29 ± 1.2) and higher AUROC (87.25 ± 0.6). C3’s across-the-board advantage thus holds even when the OOD distribution shifts between training and test.

Table 5: C3 under the hardest setting: large-scale, unseen OOD data ($\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$). Medix cuts FPR by nearly half relative to the wild-data baselines.

Method	FPR ↓	AUROC ↑
OE	68.80 ± 2.8	82.89 ± 1.1
Energy (w/ OE)	69.81 ± 2.4	85.59 ± 1.0
WOODS	69.41 ± 2.7	86.76 ± 0.8
Medix (ours)	41.29 ± 1.2	87.25 ± 0.9

A.5 Evaluating the Impact of Pseudo-Label Quality on OOD Filtering and Robustness

A further robustness question for C1 is whether the quality of the pseudo-labels $\hat{y}_{\tilde{x}_i}$ —specifically their softmax confidence—affects filtering, and whether pre-removing low-confidence pseudo-labels would help. We test this with CIFAR-10 as InD and LSUN-Resize as OOD: we compute per-sample softmax probabilities and discard predictions below a 0.6 threshold, which removes 15.98% of the training set.

The outcome is essentially a null result, and a reassuring one. Filtering and no-filtering yield the same FPR95 of 0.01%, with AUROC nudging up only marginally from 99.98% to 99.99%. Removing low-confidence pseudo-labels neither helps nor hurts, so C1 is resilient to noisy or low-confidence labels and needs no extra filtering machinery.

A.6 Evaluating Computation and Memory Efficiency in Medix Filtering

We finally ask whether C1’s filtering phase is computationally feasible and memory-efficient at scale. We profile it on an NVIDIA A100-SXM4-80GB GPU with roughly 15,000 unlabeled samples per InD–OOD pair, using CIFAR-10 and CIFAR-100 as InD and LSUN-Resize as OOD. Table 6 shows GPU memory stays modest and filtering completes in a tractable time even at thousands of samples. C1 is therefore practical, imposing no excessive memory overhead on large datasets.

Table 6: Efficiency of C1’s filtering stage on an NVIDIA A100 80GB GPU: modest peak memory and tractable wall-clock time at scale.

InD–OOD	Wall Clock Time (s)	Peak GPU Memory (MB)	Current GPU Memory (MB)
CIFAR10 – LSUN-Resize	4497.17	99.46	31.74
CIFAR100 – LSUN-Resize	5478.36	99.56	31.79

A.7 Comparison with Semi-Supervised Open-Set Recognition Methods

Our contributions differ from recent semi-supervised open-set recognition (SSL) methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) in three key ways, each tied to one of C1–C3. *Different goal.* In OpenML Saito et al. (2021), the aim is to handle outliers while training a standard classifier; in ours (C1), the aim is to detect OOD samples from unlabeled wild data and train a dedicated OOD detector. SSL methods improve classification despite outliers, whereas we improve OOD detection in an open-world setting with no labeled OOD data.

Different treatment of outliers. These SSL methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) treat outliers as noise to suppress during training, while Medix (C1) actively detects and *leverages* them to build a stronger OOD detector.

Different technique and guarantees. OpenMatch relies on consistency regularization—enforcing agreement across stochastic transformations of an input via soft constraints on classifier outputs to smooth decision boundaries. Medix instead uses median-based filtering, exploiting the median’s insensitivity to extreme values to identify OOD samples, and—uniquely—backs this with the C2 error bounds governed by the contamination effect (OOD proportion) and concentration effect (sub-Gaussian InD gradients) that keep OOD detection error low.

B Broader Impacts and Limitations

Robust OOD detection is increasingly essential to reliable machine learning in the real world. Our median-centric framework (C1), backed by guarantees (C2) and broad empirical wins (C3), improves OOD handling and model safety. The impact reaches beyond theory into practical domains—healthcare, autonomous systems, and finance—where better OOD detection addresses a key deployment challenge and fosters trust in machine learning. Promising directions include coupling C1’s filter with generative models for outlier synthesis, and adapting it to dynamic settings where the OOD distribution drifts over time.

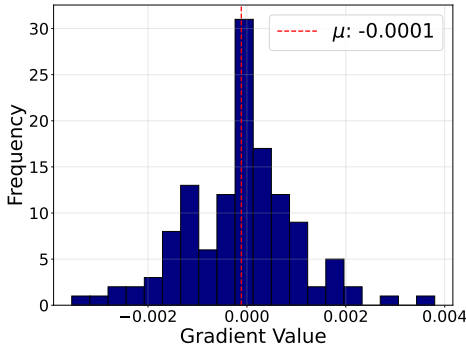
The main limitation of Medix is that C1 relies on a mixture of unlabeled InD and OOD data that may, in practice, be noisy, corrupted, or inconsistently sampled. This assumption is not unique to us—it reflects the inherent uncertainty of deployment and is a standard starting point in robust learning theory. We keep it explicit and verifiable rather than hidden: the assumptions are formalized in the main theorems of C2, and Remark 4.3 states the precise geometric separation condition needed for reliable filtering. We also extend C2 to weaker settings in Theorem C.3, dropping the sub-Gaussian requirement and showing the framework survives under milder distributional assumptions. In short, Medix’s simplifying assumptions are interpretable and relaxable, forming a principled base for future OOD detection under realistic noise.

711 C Proof of Theorem

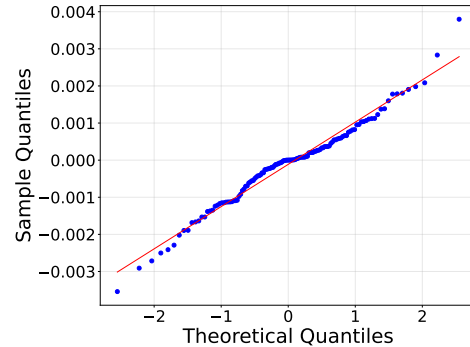
712 This appendix gives the full proofs behind contribution C2. For clarity, Table 7 first collects the
 713 key notation—gradient quantities, sample sizes, filtering terms, and the parameters entering our
 714 concentration bounds. Unless noted otherwise, vectors are boldface, $\mathcal{X}$ is the input space, and all
 715 gradients are taken with respect to a fixed pretrained model f_ϕ .

Table 7: Notation used throughout the C2 analysis. Bold symbols denote vectors and $\mathcal{X}$ the input space.

Symbol	Meaning
$\mathcal{X}$	Input space (e.g., images in $[0, 1]^d$)
f_ϕ	Model parameterized by ϕ
$\ell(f_\phi(x))$	Loss function evaluated at input x
$\nabla \ell(f_\phi(x))$	Gradient of loss at point x
$\mathbf{g}(x)$	Shorthand for $\nabla \ell(f_\phi(x))$, gradient vector
$\bar{\mathbf{g}}_{\text{in}}$	Empirical mean of gradients over InD points
m	Total number of unlabeled samples in $\mathcal{S}_{\text{wild}}$
$m_{\text{in}} = (1 - \pi)m$	Number of InD samples
$m_{\text{out}} = \pi m$	Number of OOD samples
$\pi \in (0, 1)$	OOD contamination ratio
$\mathcal{S}_{\text{wild}}$	Unlabeled mixture of InD and OOD data
$\mathcal{S}_{\text{in}}^*$	Inliers selected by the median filter
$\mathcal{S}_{\text{out}}^*$	OOD points selected by the median filter
$\mathcal{B}_\varepsilon(\bar{\mathbf{g}}_{\text{in}})$	Ball of radius ε centered at $\bar{\mathbf{g}}_{\text{in}}$
m_ε	Number of InD points inside $\mathcal{B}_\varepsilon$
m_ε^*	Number of selected inliers inside $\mathcal{B}_\varepsilon$
m_{sep}^*	Number of well-separated OODs in selected set
ERR_{in}	Fraction of InD points misclassified as OOD
ERR_{out}	Fraction of OOD points misclassified as InD
σ^2	Sub-Gaussian proxy variance of each gradient coordinate
ε	Threshold for gradient deviation, $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$
$\delta \in (0, 1)$	Confidence parameter for concentration bounds
d	Dimensionality of gradient vectors



(a) InD gradient histogram.



(b) InD gradient Q-Q plot.

Figure 4: Empirical support for C2’s sub-Gaussian assumption on InD gradient coordinates. (left) Histogram of gradient values (CIFAR-100 InD data), concentrated around the mean with light tails, consistent with sub-Gaussianity. (right) Q-Q plot of empirical InD-gradient quantiles against a Gaussian, whose close alignment confirms the assumption.

716 **C.1 Bound on the Inlier Misclassification Rate**

717 **Theorem C.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are*
 718 *i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$, and*
 719 *fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification*
 720 *rate of the element-wise median (EWM) filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

721 *Proof.* The proof of this half of C2 proceeds in two steps. **Step 1** establishes a high-probability
 722 concentration bound for InD gradients around their mean $\bar{\nabla}_{\text{in}}$, and calls an InD sample a *good inlier*
 723 when its gradient lies in this high-probability region. **Step 2** runs a “swapping” argument on the
 724 EWM objective, showing that excluding more than a $(2\eta + \frac{\pi}{2}(1-\pi))$ fraction of good inliers—where
 725 η is the fraction of InD samples whose gradients fall outside the region—would contradict optimality.
 726 Combining the two steps yields the stated bound on the inlier misclassification rate ERR_{in} .

727 **Step 1: Concentration of InD Gradients.** Let $\nabla \ell(f_{\phi}(\mathbf{x})) \in \mathbb{R}^d$ be the gradient (with respect to ϕ)
 728 of a loss evaluated at an InD sample $\mathbf{x}$. Suppose each coordinate of this gradient is sub-Gaussian with
 729 variance proxy σ^2 , centered at the same mean vector $\bar{\nabla}_{\text{in}}$. That is, for each coordinate $j \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|\nabla \ell(f_{\phi}(\mathbf{x}))_j - \bar{\nabla}_{\text{in},j}| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (6)$$

730 Define

$$\mathbf{v} := \nabla \ell(f_{\phi}(\mathbf{x})) - \bar{\nabla}_{\text{in}} \in \mathbb{R}^d.$$

731 It is straightforward to show that

$$\|\mathbf{v}\|_2 \leq \sqrt{d} \|\mathbf{v}\|_{\infty},$$

732 so if $\|\mathbf{v}\|_2 > \epsilon \sqrt{d}$ then there must exist at least one coordinate j with $|v_j| > \epsilon$.

733 Hence, combining (6) across $j = 1, \dots, d$ via a union bound yields

$$\mathbb{P}\left(\|\nabla \ell(f_{\phi}(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (7)$$

734 Let $m_{\text{in}} = (1 - \pi)m$ denote the total number of InD points in the set $\mathcal{S}_{\text{wild}}$. We define an InD point
 735 $\mathbf{x}$ to be *bad* if

$$\|\nabla \ell(f_{\phi}(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}. \quad (8)$$

736 Our goal is to show that, with high probability, only a small fraction of these m_{in} inliers can be bad.

737 From our sub-Gaussian assumption and applying the union bound across the d coordinates, we define
 738 a constant $p \in [0, 1]$ as follows:

$$p := \mathbb{P}\left(\|\nabla \ell(f_{\phi}(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (9)$$

739 Here, $\mathbf{x}$ is a generic InD sample, and we assume that the samples are i.i.d. across the m_{in} inliers.

740 Let X be the random variable denoting the number of bad inliers among the m_{in} samples. For each
 741 inlier $\mathbf{x}_i$, define the indicator variable

$$I_i := \begin{cases} 1, & \text{if } \|\nabla \ell(f_{\phi}(\mathbf{x}_i)) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

742 Then, X can be expressed as

$$X = \sum_{i=1}^{m_{\text{in}}} I_i. \quad (11)$$

743 By the linearity of expectation, we have

$$\mathbb{E}[X] = \sum_{i=1}^{m_{\text{in}}} \mathbb{E}[I_i] \quad (12)$$

$$= \sum_{i=1}^{m_{\text{in}}} \mathbb{P}(\mathbf{x}_i \text{ is bad}) \quad (13)$$

$$\stackrel{\text{from (9)}}{\leq} m_{\text{in}} \cdot p \quad (14)$$

$$\leq m_{\text{in}} \cdot 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (15)$$

744 Observing that X is a sum of m_{in} Bernoulli(p) random variables, each bounded between 0 and 1, we
745 can apply Hoeffding's inequality to obtain, for any $t > 0$,

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (16)$$

746 Setting

$$t := \sqrt{\frac{m_{\text{in}} \log\left(\frac{1}{\delta}\right)}{2}}, \quad (17)$$

747 we obtain

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (18)$$

748 With probability at least $1 - \delta$, we therefore have

$$X \leq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (19)$$

749 Dividing both sides by m_{in} yields an upper bound on the fraction of bad inliers:

$$\frac{X}{m_{\text{in}}} \leq \frac{\mathbb{E}[X]}{m_{\text{in}}} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (20)$$

$$\stackrel{\text{from (12)}}{\leq} 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (21)$$

$$= \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}. \quad (22)$$

750 Let us denote this fraction by η . By setting $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$ Thus, with probability at least
751 $1 - \delta$, no more than an $O\left(d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{m_{\text{in}}}}\right)$ fraction of inliers have gradients lying
752 outside the radius $\epsilon\sqrt{d}$ around $\bar{\nabla}_{\text{in}}$. These are precisely the *bad inliers*, while the remainder are
753 called *good inliers*.

754 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
755 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (23)$$

756 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
757 thereby contradicting the optimality of $\mathcal{S}^*$.

758 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 759 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (24)$$

760 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

761 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (25)$$

762 Define the swap size as

$$m'_\varepsilon := \min \{ m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}} \}. \quad (26)$$

763 This quantity counts how many good inliers (those inside B_ε) are currently excluded from $\mathcal{S}^*$ and
 764 available for swapping in.

765 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 766 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

767 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (27)$$

768 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (28)$$

769 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (29)$$

770 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (30)$$

771 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 772 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (31)$$

773 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (32)$$

774 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (33)$$

775 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (34)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (35)$$

776 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (36)$$

777 so the points lying inside B_ε form a strict majority in $\widehat{\mathcal{S}}$.

778 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
779 gradients in $\mathcal{S}^*$ and $\widehat{X}^{(j)} = \{\widehat{x}_1^{(j)}, \dots, \widehat{x}_m^{(j)}\}$ the same in $\widehat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

780 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
781 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
782 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\widehat{X}^{(j^*)}) \in I_{j^*}, \quad (37)$$

783 which implies

$$|\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (38)$$

784 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
785 increase it, so the median distance from μ_j does not increase and possibly decreases.

786 Thus, component-wise:

$$|\text{EWM}_j(G_{\widehat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (39)$$

787 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\widehat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (40)$$

788 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
789 the proof.

$$\Rightarrow \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (41)$$

790 □

791 C.2 Bound on the Outlier Retention Rate

792 We now prove the second half of C2: a non-asymptotic upper bound on the fraction of OOD points
793 retained by the MEDIX filter. The argument mirrors the inlier bound (Appendix C.1), but replaces
794 inlier concentration with a separation assumption between the OOD and InD gradient means.

795 **Theorem C.2** (Outlier Retention Bound under Vector Separation). *Assume that for every OOD point*
796 *$x \sim P_{\text{out}}$, the gradient $g(x) := \nabla \ell(f_{\varphi_{\text{in}}}(x)) \in \mathbb{R}^d$ satisfies:*

797 **(I) Sub-Gaussianity.** *Each coordinate $g_j(x)$ is sub-Gaussian with variance proxy σ_{out}^2 .*

798 **(II) Vector Mean Separation.** *The mean OOD gradient vector μ_{out} satisfies*

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta \sqrt{d}, \quad \text{for some } \Delta > 0.$$

799 Fix any tolerance $\varepsilon \in (0, \Delta)$ and confidence parameter $\delta \in (0, 1)$. Define:

$$p_{\text{out}}(\varepsilon) := 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right), \quad \eta_{\text{out}} := p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}.$$

800 Then with probability at least $1 - \delta$, the outlier retention rate satisfies:

$$\text{ERR}_{\text{out}} := \frac{|\mathcal{S}_{\text{in}}^* \cap \text{OOD}|}{m_{\text{out}}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}. \quad (42)$$

801 *Proof. Step 1: Concentration of OOD Gradients.*

802 By assumption, each coordinate of $g(x)$ is sub-Gaussian, and the mean vector is separated from $\bar{\nabla}_{\text{in}}$
803 by at least $\Delta\sqrt{d}$. Therefore, for any OOD point x , the event

$$\|g(x) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \quad (43)$$

804 has probability at most $p_{\text{out}}(\varepsilon)$.

805 **Counting the “ambiguous” OOD points.** For every OOD example x_i define the indicator

$$J_i := \mathbf{1}\left\{\|g(x_i) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}, \quad i = 1, \dots, m_{\text{out}}.$$

806 Each $J_i \sim \text{Bernoulli}(p_{\text{out}}(\varepsilon))$ and the J_i ’s are independent. Let

$$Y := \sum_{i=1}^{m_{\text{out}}} J_i, \quad \mu := \mathbb{E}[Y] = m_{\text{out}} p_{\text{out}}(\varepsilon).$$

807 Because the indicators J_i take values in $\{0, 1\}$, Hoeffding’s inequality states that for any $s > 0$

$$\Pr\{Y - \mu \geq s\} \leq \exp\left(-\frac{2s^2}{m_{\text{out}}}\right).$$

808 Choose $s = \sqrt{\frac{m_{\text{out}} \log(1/\delta)}{2}}$; then (C.2) equals δ , and with probability at least $1 - \delta$

$$Y \leq m_{\text{out}} \left(p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}\right) = m_{\text{out}} \eta_{\text{out}}.$$

809 Define the high-probability event

$$\mathcal{E}_1 := \{Y \leq m_{\text{out}} \eta_{\text{out}}\}, \quad (35)$$

810 which holds with probability at least $1 - \delta$.

811 **Step 2: Contradiction via a Swapping Argument** Suppose, toward contradiction, that

$$\text{ERR}_{\text{out}} > \eta_{\text{out}} + \frac{1 - \pi}{2\pi}. \quad (44)$$

812 Let $m_{\text{out}}^* := |\mathcal{S}_{\text{in}}^* \cap \text{OOD}|$, and define $m_{\varepsilon}^* := \#\{\text{ambiguous OODs in } \mathcal{S}_{\text{in}}^*\}$. Then, under $\mathcal{E}_1$,

$$m_{\text{sep}}^* := m_{\text{out}}^* - m_{\varepsilon}^* > \frac{1 - \pi}{2\pi} m_{\text{out}} = \frac{1}{2} m_{\text{in}}. \quad (45)$$

813 Thus, more than half the points in $\mathcal{S}_{\text{in}}^*$ are separated OODs whose gradients lie outside the ball

$$B_{\varepsilon} := \left\{g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}. \quad (46)$$

814 By the definition of the coordinate-wise median, this implies that in every coordinate, the majority of
815 entries in $G_{\mathcal{S}_{\text{in}}^*}^{(j)}$ lie outside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, hence:

$$\|\text{EWM}(G_{\mathcal{S}_{\text{in}}^*}) - \bar{\nabla}_{\text{in}}\|_2^2 \geq \varepsilon^2 d. \quad (47)$$

Now consider a new subset $\hat{\mathcal{S}}$ formed by swapping the m_{sep}^* separated OODs in $\mathcal{S}_{\text{in}}^*$ with an equal number of InD points whose gradients lie inside B_ε . Such inliers exist because fewer than $m_{\text{out}}\eta_{\text{out}}$ are ambiguous, and the total number of InD points is m_{in} .

Then at least half the entries in every coordinate of $G_{\hat{\mathcal{S}}}^{(j)}$ lie inside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, implying:

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2^2 \leq \varepsilon^2 d. \quad (48)$$

Combining with (47) yields a contradiction:

$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\hat{\mathcal{S}}})\|_2^2 < \|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}_{\text{in}}^*})\|_2^2. \quad (49)$$

This contradicts the optimality of $\mathcal{S}_{\text{in}}^*$, and so the assumption must be false. Therefore, our assumption must be false. With probability at least $1 - \delta$, we conclude that the outlier retention rate is bounded by

$$\text{ERR}_{\text{out}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}, \quad (50)$$

□

C.3 Theorem C.3 without Sub-Gaussianity

Theorem C.3 (Inlier Misclassification Bound without Sub-Gaussianity). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and that each coordinate has variance σ^2 and finite fourth moment bounded by μ_4 , i.e.,*

$$\mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^2 = \sigma^2, \quad \mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^4 \leq \mu_4.$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the number of InD samples in $\mathcal{S}_{\text{wild}}$, and fix any tolerance $\varepsilon > \sigma$ and confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the element-wise median (EWM) filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{2 \left(\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \right)}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1 - \pi)}}_{\text{Contamination term}}.$$

What relaxing the assumption costs. This theorem shows C2 does not hinge on sub-Gaussianity; here we spell out the price of dropping it. The original, sub-Gaussian bound carries a decay term

$$2 \exp\left(-\frac{\varepsilon^2 d}{2\sigma_{\text{sub}}^2}\right)$$

that shrinks exponentially in the dimension d , making it extremely tight for large d . The new bound, assuming only finite fourth moments, instead carries

$$\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2}$$

which decays merely as $O(1/d)$ —looser for large d , but valid under heavier tails. Reassuringly, both bounds keep the same finite-sample term

$$\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}},$$

which shrinks with m_{in} regardless of the tail assumption. So the core robustness message of C2 persists even without sub-Gaussian tails.

841 *Proof.* The proof mirrors the two-step structure of the original Theorem 4.1: (1) bound the fraction
 842 of “bad” InD samples (those far from ∇_{in}), now *without* sub-Gaussianity, and (2) apply the same
 843 swapping argument to bound the inlier misclassification rate. The only substantive change is replacing
 844 the sub-Gaussian tail bound with a moment-based bound from the finite fourth-moment assumption.

845 **Step 1: Concentration of InD Gradients** Define the gradient deviation for an InD sample x as:

$$v = \nabla \ell(f_\phi(x)) - \nabla_{\text{in}} \in \mathbb{R}^d, \quad (51)$$

846 where $\nabla_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in S_{\text{in}}} \nabla \ell(f_\phi(x_i))$ is the mean gradient over the labeled InD training set S_{in} ,
 847 and we assume ∇_{in} is the population mean. An InD sample is classified as “bad” if:

$$\|v\|_2 > \epsilon \sqrt{d}, \quad (52)$$

848 and “good” otherwise. Our goal is to bound the fraction of bad inliers among the m_{in} InD samples in
 849 S_{wild} .

850 Since $\|v\|_2^2 = \sum_{j=1}^d v_j^2$, we analyze the ℓ_2 norm via the sum of squared coordinates. Assume that
 851 the coordinates $v_j = \nabla \ell(f_\phi(x))_j - \nabla_{\text{in},j}$ are independent across $j = 1, \dots, d$, each with:

$$\mathbb{E}[v_j] = 0 \quad (\text{assuming } \nabla_{\text{in},j} = \mathbb{E}[\nabla \ell(f_\phi(x))_j]), \quad (53)$$

$$\mathbb{E}[v_j^2] = \sigma^2, \quad \mathbb{E}[v_j^4] \leq \mu_4. \quad (54)$$

852 Thus:

$$\mathbb{E}[\|v\|_2^2] = \mathbb{E}\left[\sum_{j=1}^d v_j^2\right] = d\sigma^2, \quad (55)$$

$$\text{Var}(\|v\|_2^2) = \text{Var}\left(\sum_{j=1}^d v_j^2\right) = \sum_{j=1}^d \text{Var}(v_j^2). \quad (56)$$

853 Compute the variance of v_j^2 :

$$\text{Var}(v_j^2) = \mathbb{E}[v_j^4] - (\mathbb{E}[v_j^2])^2 \leq \mu_4 - \sigma^4, \quad (57)$$

854 so:

$$\text{Var}(\|v\|_2^2) \leq d(\mu_4 - \sigma^4). \quad (58)$$

855 Using Chebyshev’s inequality for $\|v\|_2^2$:

$$\mathbb{P}(|\|v\|_2^2 - d\sigma^2| \geq t) \leq \frac{d(\mu_4 - \sigma^4)}{t^2}. \quad (59)$$

856 We need $\mathbb{P}(\|v\|_2 > \epsilon \sqrt{d})$, which corresponds to $\|v\|_2^2 > \epsilon^2 d$. Set the threshold:

$$\|v\|_2^2 > \epsilon^2 d \implies \|v\|_2^2 - d\sigma^2 > \epsilon^2 d - d\sigma^2 = d(\epsilon^2 - \sigma^2). \quad (60)$$

857 Thus:

$$\mathbb{P}(\|v\|_2^2 > \epsilon^2 d) = \mathbb{P}(\|v\|_2^2 - d\sigma^2 > d(\epsilon^2 - \sigma^2)) \leq \frac{d(\mu_4 - \sigma^4)}{[d(\epsilon^2 - \sigma^2)]^2} = \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}, \quad (61)$$

858 provided $\epsilon > \sigma$ so that $\epsilon^2 - \sigma^2 > 0$. Define:

$$p = \mathbb{P}\left(\|\nabla \ell(f_\phi(x)) - \nabla_{\text{in}}\|_2 > \epsilon\sqrt{d}\right) \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (62)$$

859 Let X be the number of bad inliers among the m_{in} InD samples in S_{wild} , where each InD sample is
860 bad with probability at most p . Then:

$$\mathbb{E}[X] \leq m_{\text{in}}p \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (63)$$

861 Since $X = \sum_{i=1}^{m_{\text{in}}} I_i$, where $I_i = 1$ if the i -th InD sample is bad and 0 otherwise, and I_i are i.i.d.
862 Bernoulli(p) variables bounded in $[0, 1]$, apply Hoeffding's inequality:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (64)$$

863 Set $t = \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}$, so:

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (65)$$

864 With probability at least $1 - \delta$:

$$X \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (66)$$

865 The fraction of bad inliers is:

$$\eta = \frac{X}{m_{\text{in}}} \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}. \quad (67)$$

866 This η represents the upper bound on the fraction of InD samples whose gradients deviate from ∇_{in}
867 by more than $\epsilon\sqrt{d}$, relying only on finite fourth moments and coordinate independence.

868 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if S^* excludes an excessive
869 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (68)$$

870 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 871 thereby contradicting the optimality of $\mathcal{S}^*$.

872 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 873 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (69)$$

874 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

875 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (70)$$

876 Define the swap size as

$$m'_\varepsilon := \min \{m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (71)$$

877 This quantity counts how many good inliers (those inside B_ε) are currently excluded from $\mathcal{S}^*$ and
 878 available for swapping in.

879 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 880 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

881 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (72)$$

882 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (73)$$

883 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (74)$$

884 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (75)$$

885 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 886 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (76)$$

887 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (77)$$

888 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (78)$$

889 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (79)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (80)$$

890 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (81)$$

891 so the points lying inside B_ε form a strict majority in $\widehat{\mathcal{S}}$.

892 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
893 gradients in $\mathcal{S}^*$ and $\widehat{X}^{(j)} = \{\widehat{x}_1^{(j)}, \dots, \widehat{x}_m^{(j)}\}$ the same in $\widehat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

894 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
895 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
896 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\widehat{X}^{(j^*)}) \in I_{j^*}, \quad (82)$$

897 which implies

$$|\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (83)$$

898 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
899 increase it, so the median distance from μ_j does not increase and possibly decreases.

900 Thus, component-wise:

$$|\text{EWM}_j(G_{\widehat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (84)$$

901 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\widehat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (85)$$

902 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
903 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (86)$$

904

□